

Optimizer Readiness Audit

Information-only assessment of whether the current ECR-2 simulator is ready to support a preliminary optimizer.

Workspace: LLX-RND-2026-0003
Selected simulator run: 656
Persisted time: 23 August 2026, 14:13:10.713 UTC
Local time: 23 August 2026, 19:43:10.713 IST
Engine: llx-ecr-simulator 2.1.0 | **Revision:** 8

FINAL CLASSIFICATION: NOT READY FOR PRELIMINARY OPTIMIZER

Purpose. Record the exact persisted state and distinguish numerical convergence from physical, thermodynamic, correlation, and governance readiness.

Change control. No code, design inputs, UI, database, solver, governance records, or optimizer task were changed. No new simulations were run for this report.

Prepared from persisted run snapshots and repository governance/evidence records. This report is not a release-grade equipment design basis.

1. Scope and selected evidence

This audit is pinned to the user-specified evening run. The selected run is not replaced by later simulator executions.

Evidence item	Use in this audit
Run 656 input/result snapshot	Primary current-run evidence.
Revision 8 C2 process-design result	Thermodynamic cascade, target, and theoretical-stage provenance.
Runs 631, 639, 643	Persisted comparative ECR-2 simulator cases.
ECR-2 BVP and correlation governance	Acceptance gates, fail-closed behavior, and applicability limits.
Existing repository tests	Mechanics/regression evidence only; not a grid-independence or pilot-validation substitute.

Workspace distinction. The database contains LLX workspaces 0001, 0002, and 0003. Only 0003 has ECR-2 simulator runs. The “three cases” are three runs inside 0003, not three independently persisted ECR-2 workspaces.

2. Run identity and status

CALCULATION STATUS warning 10 warnings; 0 validation issues	BVP CONVERGENCE Accepted 13 iterations; four acceptance gates passed
FINAL NORMALIZED RESIDUAL 6.35e-14 Dimensionless residual measure	TOTAL MASS-BALANCE ERROR 1.29e-10 kg/h Internal numerical closure

The four independent BVP gates are normalized residual, relative state change, maximum component balance, and total mass balance. Passing them demonstrates numerical closure of the implemented equations; it does not validate the RRBO/NMP physical model.

3. Persisted BVP input vector

Feed and operating basis

Input	Value	Basis/status
RRBO feed	3,396 kg/h	Physical feed
NMP feed	3,952 kg/h	Physical feed
S/O	1.0 volume basis; 1.163722 mass basis	Input conversion
NMP purity	0.995	Physical plant-boundary input; not C2 solvent coordinate
Extraction / operating temperature	70 °C / 343.15 K	Thermodynamic evaluation temperature
Feed temperature anchor	25 °C	Persisted feed-side anchor
Feed / extraction / design pressure	2.5 / 1.0 / 2.5 bar	Pressure inputs
Phase configuration	NMP continuous / RRBO dispersed	Supported orientation

Physical RRBO composition

Component	Mass fraction	C2 normalized mole fraction
Saturates	0.750000	0.71177
Mono-aromatics	0.083333	0.12689
Di-aromatics	0.083333	0.09474
Poly-aromatics	0.083333	0.06661
NMP	0	0

The C2 feed aromatic content is **28.82286 mol%**. The C2 target is **10 mol% total aromatics** on the five-component thermodynamic basis.

4. Properties, partition, and thermodynamic basis

Property / contract	Persisted value	Evidence status
RRBO density	849 kg/m ³ at 70 °C	EPD tabular; Assumed/provisional
RRBO viscosity	0.0598 Pa·s; reference 40 °C	Auto-populated; not a validated 70 °C measurement
NMP viscosity	0.000917 Pa·s at 70 °C	Assumed
Interfacial tension	0.010 N/m at 70 °C	Vendor source
RRBO sulfur	1,500 ppm input	Not modeled in ECR-2
Partition basis	K _d concentration	Engineer-approved governed basis
Partition definition	$K_{d,i} = C_{d,i} / C_{c,i}$	Local NRTL equilibrium target
NRTL	$\alpha = 0.2$; $\tau = b/T$	Calibrated at 298.15 K only
Evaluation temperature	343.15 K	44.50 K outside calibrated range

The revision-8 C2 record reports a fractional stage count of **0.82802**, rounded to **$N_T = 1$** . At one C2 cascade stage, the reported raffinate aromatic content is 6.0906 mol%, below the 10 mol% target.

Thermodynamic limitation. NRTL calibration reproduction at 298.15 K has maximum deviation 0.092 against a 0.009 gate, with 2 of 13 tie-lines within the gate. At 70 °C the result is explicitly “Temperature Extrapolation — Preliminary / Pending Validation.”

5. Diffusivities and physical molecular weights

Component	Continuous phase (m ² /s)	Dispersed phase (m ² /s)	Classification
Saturates	4.28373e-10	4.55160e-11	Assumed/project
Mono-aromatics	4.00780e-10	4.25842e-11	Assumed/project
Di-aromatics	3.72830e-10	3.96144e-11	Assumed/project
Poly-aromatics	3.40076e-10	3.61341e-11	Assumed/project
NMP	8.03358e-10	9.28448e-11	Assumed/project

Physical component	MW (g/mol)	Status
Saturates	269.93	Assumed/auto-basis
Mono-aromatics	320.00	Assumed/auto-basis
Di-aromatics	377.57	Assumed/auto-basis
Poly-aromatics	459.45	Assumed/auto-basis
NMP	99.13	Physical solvent basis

These physical/project values are separate from the Coto/NRTL surrogate compounds. They remain active preliminary inputs and are not evidence of validated RRBO/NMP transport properties.

6. Geometry and hydrodynamic inputs

Input	Value
Column diameter	0.85 m
Active height	0.75 m
Numerical compartments / compartment height	3 / 0.25 m
Rotor diameter	0.425 m
Rotor/column diameter ratio	0.5
Rotor speed	60 rpm
Rotor label	Kühni turbine
Rotors per compartment	1
Stator open-area fraction	0.4
Power density / dissipation basis	0.097741 W/kg
Power-number basis	0.39

Discretization rule. The three numerical compartments are a solver grid, not a free equipment-design variable. Treating them as an optimizer variable would create a numerical rather than physical optimum.

7. d₃₂, holdup, and interfacial area

The direct-turbulence route is:

$$\varepsilon = N_e \, n^3 \, d_R^5 \, / \, V_R$$

$$d_{32} = C \, (\gamma \, / \, \rho_c)^{0.6} \, \varepsilon^{-0.4}$$

Output	Run-656 result	Status
d ₃₂	0.995817 mm	DIRECT_TURBULENCE_D32_PRELIMINARY
C	0.39	Project-controlled midpoint of 0.36–0.43; not pilot validated
Dispersed holdup, φ _d	0.204662	Preliminary correlation
Interfacial area, a	1,233.128 m ² /m ³	Computed from d ₃₂ and holdup

The local area uses $a = 6 \, \phi_d \, / \, d_{32}$, and the BVP-local area records match that equation. This is an internal consistency result, not experimental validation.

8. Run-656 outlet results

Raffinate

Component	Flow (kg/h)	Mass fraction
Saturates	1,931.598	0.796908
Mono-aromatics	101.355	0.041816
Di-aromatics	104.088	0.042943
Poly-aromatics	144.966	0.059808
NMP	141.858	0.058526
Total	2,423.865	1.000000

Extract

Component	Flow (kg/h)	Mass fraction
Saturates	615.405	0.125481
Mono-aromatics	181.644	0.037037
Di-aromatics	178.911	0.036480
Poly-aromatics	138.033	0.028145
NMP	3,790.382	0.772857
Total	4,904.375	1.000000

9. Recovery and product-quality interpretation

Derived quantity	Result	Definition
RRBO to raffinate	2,282.007 kg/h; 67.197%	All four RRBO classes in raffinate / RRBO feed
RRBO to extract	1,113.993 kg/h; 32.803%	All four RRBO classes in extract / RRBO feed
Saturates recovery	56.879%	Raffinate saturates / feed saturates
Aromatic extraction yield	Approximately 58.73%	Extract mono+di+poly / feed mono+di+poly
Raffinate aromatics, including NMP	9.562 mol%	Physical MW calculation on total raffinate phase
Hydrocarbon-only raffinate aromatics	Approximately 11.3 mol%	NMP excluded from the denominator

NMP is excluded from the aromatic recovery numerator and denominator. The 10 mol% C2 target is on a five-component thermodynamic basis, whereas a commercial hydrocarbon product specification may exclude residual NMP. The product-quality constraint therefore needs an explicit basis before optimization.

Sulfur/DBT: Not yet available. A sulfur input exists, but no ECR-2 sulfur/DBT transfer and product-quality model is executed.

10. Output classification

Output	Classification	Audit interpretation
BVP residual, component balance, total balance	VERIFIED internally	Numerical acceptance only.
C2 fractional/rounded N_T	PRELIMINARY	NRTL extrapolation and calibration mismatch.
Local NRTL equilibrium	PRELIMINARY	70 °C extrapolation from 25 °C calibration.
d_{32}	PRELIMINARY	Direct-turbulence route not pilot validated.
Holdup and interfacial area	PRELIMINARY	Dependent on preliminary applicability basis.
Reynolds/Schmidt/Sh/film coefficients	PRELIMINARY / restricted	Property and secondary-correlation limitations.
$K_{overall}$, K_{oa} , component rates	PRELIMINARY / restricted	Not release/design-decision eligible.
Outlet flows and derived recovery	PRELIMINARY	Forward BVP output on preliminary physical basis.
Sulfur/DBT quality	NOT YET AVAILABLE	No model.
Flooding, economics, CAPEX, energy objective	NOT IMPLEMENTED	Separate or future models required.

11. C1/C2 correlation governance

Constant/route	Value/status	Optimizer treatment
Project-provisional C1	0.90; role unresolved; not applied numerically	Never a decision variable
Project-provisional C2	0.45; role unresolved; not applied numerically	Never a decision variable
Scoped Kühni secondary C1	7.5; separately registered for preliminary equation	Fixed governed metadata only
Kühni C2 and placement	Unresolved for release use	Cannot support optimizer ranking

Project-provisional values, secondary literature metadata, and governed release correlations must remain separate. A numerically available constant is not automatically a validated optimization input.

12. Legitimate variables and bounds

Candidate variable	Current value	Governed bounds	Status
Temperature	70 °C	Not governed	Blocked for optimization
S/O ratio	1.0 volume basis	Not governed	Numerically perturbable only
Rotor speed	60 rpm	Not governed	Preliminary only
Active height	0.75 m	Not governed	Needs grid/transfer validation
Column diameter	0.85 m	Not governed	Scale-up basis absent
Rotor ratio	0.5	Not governed	Engineering input, not an approved bound
d ₃₂ C coefficient	0.39	0.36–0.43 sensitivity range	Not an optimization variable
Numerical compartments	3	Solver choice	Must be excluded from design optimization

13. Objective and temperature treatment

No governed optimizer objective currently exists. Possible future objectives—minimize height, diameter, energy, or RRBO loss while meeting aromatic and sulfur specifications—remain candidate concepts only.

An optimizer cannot safely rank run 656 outputs because the physical transfer response is preliminary, sulfur is unavailable, no flooding limit is implemented, no economic model is persisted, and the product-quality basis is not fully resolved.

Temperature conclusion. Temperature is not optimizer-ready. The 70 °C equilibrium result relies on NRTL extrapolation from a 25 °C calibration and is explicitly pending validation against controlled RRBO/NMP LLE data near the operating temperature.

The C2 theoretical stage result must remain separate from ECR-2 physical compartment semantics. C2 fractional $N_T = 0.82802$, rounded $N_T = 1$, does not mean that the ECR-2 column has one physical compartment or one validated transfer stage.

14. Grid independence

Grid independence is **not demonstrated**.

- Run 656 uses three numerical compartments.
- No persisted same-operating-point comparison across refined compartment grids was found.
- Existing tests verify BVP mechanics, balances, and regression behavior.
- Those tests do not establish outlet, recovery, or objective independence from compartment count.

A future grid audit must compare the same operating point across multiple compartment counts, including outlet compositions, recovery, transfer, d_{32} /holdup profiles, residuals, and all objective-relevant quantities.

15. Persisted comparison cases

Run	S/O	D (m)	d _R (m)	d ₃₂ (mm)	φ _d	a (m ² /m ³)
631	2.0	1.05	0.525	0.773	0.2700	2,096.7
639	1.0	0.85	0.425	0.996	0.2047	1,233.1
643	0.5	0.75	0.375	1.157	0.1651	855.8
656	1.0	0.85	0.425	0.996	0.2047	1,233.1

Runs 639 and 656 have the same numerical operating point and identical result snapshot. Runs 631, 639, and 643 all use 0.75 m height, three compartments, 60 rpm, and a 0.5 rotor ratio.

The comparison is not a one-variable S/O sensitivity: diameter and rotor diameter changed with S/O. At fixed rotor ratio, the observed trend is consistent with approximately $\varepsilon \propto D^3$, so reduced diameter reduces dissipation and increases d₃₂.

16. Sensitivity readiness

Variable	Can calculate a perturbation?	Readiness	Limitation
Temperature	Yes	Not ready	NRTL temperature extrapolation
S/O	Yes	Partial numerical evidence only	Existing cases confounded with diameter; no governed bounds
Rotor speed	Yes	Not ready	d ₃₂ and transfer correlations preliminary
Active height	Yes	Not ready	Grid independence and transfer validation absent
Column diameter	Yes	Not ready	Geometry/d ₃₂ coupling and scale-up basis absent

Calculation capability is not the same as optimizer readiness. A variable is not ready until its model response, bounds, constraints, and evidence status can be audited independently.

17. Fail-closed conditions

The optimizer must represent the following as **INFEASIBLE**, never as merely poor objective values:

- Unsupported phase orientation
- Missing/unapproved C2 basis
- Missing partition basis
- Missing d_{32} configuration
- Invalid flow vector
- Invalid physical MWs
- Invalid local composition
- Failed property closure

- Failed NRTL flash
- Invalid holdup or d_{32}
- Invalid interfacial area/slip
- Invalid transfer volume
- Donor-inventory violation
- Negative/non-finite state
- BVP non-convergence
- Failed component/total balance

Missing evidence, out-of-scope correlation applicability, and an unavailable requested sulfur/DBT constraint must also produce a hard infeasibility state.

18. Optimizer-exploit audit and validation requirements

Principal exploit risks

1. Lower diameter could improve apparent d_{32} through the preliminary turbulence equation.
2. Higher S/O could appear beneficial without a solvent-cost, capacity, or flooding constraint.
3. The preliminary d_{32} coefficient range could be exploited as a hidden decision variable.
4. $C2 N_T = 1$ could be misread as an ECR-2 physical stage count.
5. Numerical compartment count could be varied to create a numerical optimum.
6. Assumed MWs, diffusivities, and viscosities could be treated as measured facts.
7. Numerical convergence could be mistaken for validated transfer performance.
8. Sulfur compliance could be claimed even though sulfur/DBT is not modeled.

Required validation before optimizer use

- Controlled RRBO/NMP tie-lines at or near 70 °C, including water and purity effects.
- Validation of local partition and NRTL behavior against measured system data.
- Pilot d_{32} measurements versus rpm, rotor/column geometry, power, and phase loading.
- Holdup, flooding margin, residence-time, and axial-mixing validation.
- Component mass-transfer coefficient and rate measurements.
- Temperature-dependent density, viscosity, interfacial tension, MW, and diffusivity evidence.
- Sulfur/DBT extraction and product-quality model validation.
- Grid-independence evidence using persisted same-point refined-grid comparisons.

19. Final readiness matrix

Requirement	Result
BVP numerical convergence	Pass, internal only
Component and total balances	Pass, internal only
Governed thermodynamics at operating temperature	Fail
RRBO/NMP equilibrium validation	Fail
Validated d_{32} , holdup, interfacial area	Fail
Validated mass-transfer coefficients/rates	Fail
Product-quality basis fully defined	Partial
Sulfur/DBT constraint enforcement	Not available
Governed optimizer variable bounds	Not established
Governed objective function	Not established
Temperature, S/O, rpm, height, diameter sensitivity readiness	Not ready
Grid independence	Not demonstrated
Optimizer fail-closed behavior	Required; optimizer not implemented
Independent pilot/industrial validation	Not available

NOT READY FOR PRELIMINARY OPTIMIZER

Run 656 remains useful as a numerically converged, preliminary engineering simulation snapshot. It is not a validated optimization kernel or release/design-decision basis.